Computing isogenies through graph theory: faster, simpler, broader

Max Duparc



Arithmetic Seminar in Lyon: October 22, 2026

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Today's program

1. Introduce a simple theory of isogeny computing in Mumford's theta model[a].
2. It naturally provides the simplest and fastest generic isogeny formulae.
3. This theory leads to further speedups on special varieties.

[a] Focus on 2-isogenies.

On the shoulders of giants

Talk based on the following works:

- [DD26]: *Chasing Rabbits Through Hypercubes: Better algorithms for higher dimensional 2-isogeny computations*, with Pierrick Dartois, ANTS 2026
- [BDD+26]: *MIKE: a fast and compact post-quantum NIKE*, with the MIKE-team, eprint.
- *Why do it simply when you can do complex? Isogeny edition*, with Giacomo Pope; Krijn Re?nders (ongoing)

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$$f : (E_1, e^{E_1}) \longrightarrow (E_2, e^{E_2})$$

- Surjective morphism $\implies |\ker(f)| < \infty$.
- Compatible with group structures.
- $\ker(f)$ cyclic.

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- dual $\widetilde{f} : (E_2, e^{E_2}) \to (E_1, e^{E_1})$ with $\ker(\widetilde{f}) = f\,(E_1[\deg(f)])$.

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- $\ker(f)$ max. isotropic ($e^A|_{\ker(f)^2} \equiv 1$).

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- **Symplectic representation** of $A[N]$:

$$\mu : (A[N], e_A) \xrightarrow{\cong} (\mathbb{Z}_N^g \times \widehat{\mathbb{Z}_N^g}, \epsilon)$$

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- Equivalent to **symplectic basis** $\mathcal{B}_\mu := \{S_1, \ldots, S_g, T_1, \ldots, T_g\}$:

$$e_A(S_i, S_j) = e_A(T_i, T_j) = 1, \quad e_A(S_i, T_j) = \zeta^{\delta_{i,j}}$$

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satisfying the **theta group action relation**:

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  - Then it is *symmetric*: (i.e. $\theta^A(-P) = \theta^A(P)$).
- [Mum66]: Uniquely determined by a 4-symplectic basis.

Properties of theta structure

- All infos[1] about A are in $\theta^A(0_A)$ the **theta-null point**.



[1] Note: this is only conjectured for finite field and $g \geq 4$.

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- Duality

$$A[4] = \{S_1, \dots, S_g, T_1, \dots, T_g\} \qquad A[4] = \{T_1, \dots, T_g, -S_1, \dots, -S_g\}$$



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The theta arithmetic is beautiful

Notation:

$$(x_i)_i \odot (y_i)_i = (x_i y_i)_i$$
$$(x_i)_i^{\odot n} = (x_i^n)_i \text{ for } n \in \mathbb{Z}.$$

$$\mathcal{H} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$ Hadamard transform

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[Gau07] `Double` : $\theta^A(P) \mapsto \theta^A([2]P)$

1. $Q \leftarrow P^{\odot 2}$
2. $Q \leftarrow \mathcal{H}(Q)$
3. $Q \leftarrow Q^{\odot 2}$
4. $Q \leftarrow Q \odot C_1$
5. $Q \leftarrow \mathcal{H}(Q)$
6. **return** $Q \odot C_2$

C_1, C_2 are derived from $\theta^A(0)$

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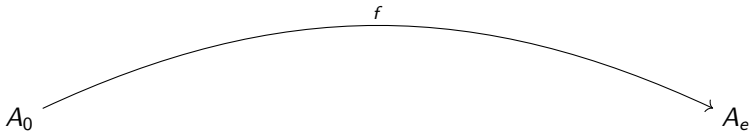
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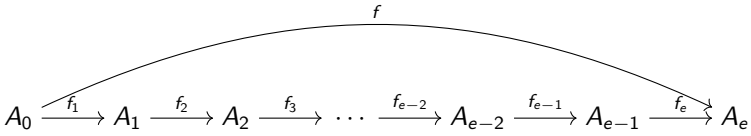
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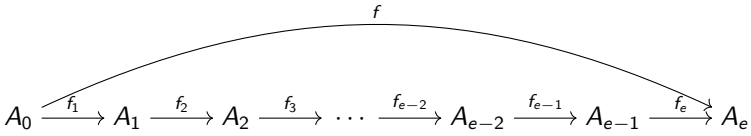
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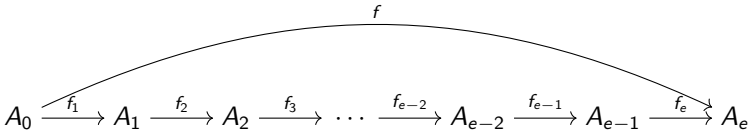
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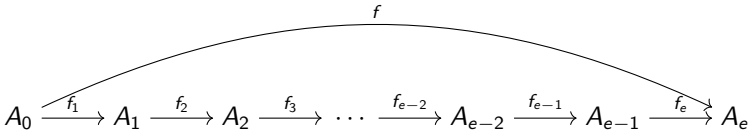


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- Bottleneck: How do we compute $\widetilde{\theta}^B(0_B)^{\odot -1}$?

Radical formulae

- Can be computed using sqrts.

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- BUT:
  - sqrts are expensive.
  - No control on the codomain theta structure.
  - For $g \geq 3$, not all choices are valid. [KMM+25]

- Not a good constructive choice.

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Interpolation formulae

A solution of the equation systems

$$\left\{ \widetilde{\theta}^B_j(f(T_{\mathbf{k}})) \cdot \widetilde{\theta}^B_j(0_B) = \alpha_{j,\mathbf{k}} \text{ s.t. } \mathbf{k} \in S \subseteq \mathbb{Z}_2^g, j \in [2^g] \right\}$$

IIIP

Let:

- $S \subseteq \mathbb{Z}_2^g$ be the *support*.
- $(\pi_{i,j})_{i \in \mathbb{Z}_2^g, j \in S}$ be such:

$$\forall i \in \mathbb{Z}_2^g, j \in S, \pi_{i,j} := \widetilde{\theta}_i^B(0) \cdot \widetilde{\theta}_i^B(f(T_j))$$

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[DD26] IIIP theorem (informal)

It's just graph topology !

IIIP as Graph Theory

$\pi_{i,j} \rightsquigarrow G := (V, E)$ codomain graph

- x_i are vertices.

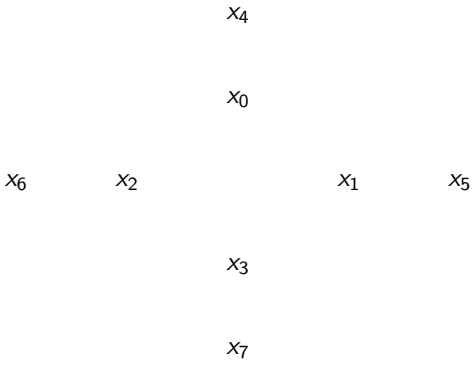


Figure: Codomain Graph G for $g = 3$, $S = \{\quad\}$

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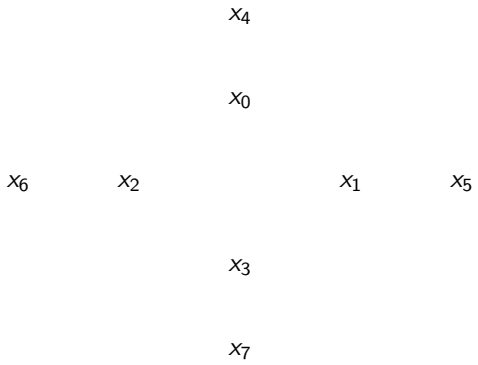


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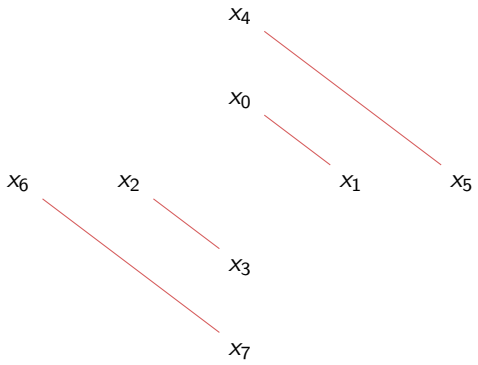


Figure: Codomain Graph G for $g = 3$, $S = \{1 \quad \}$

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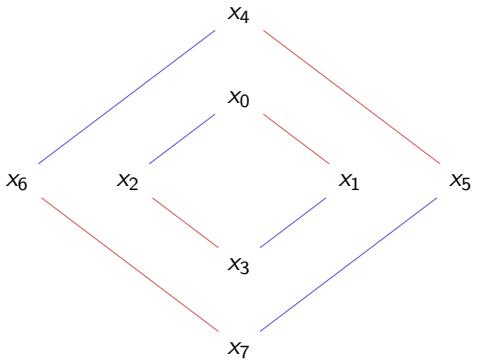


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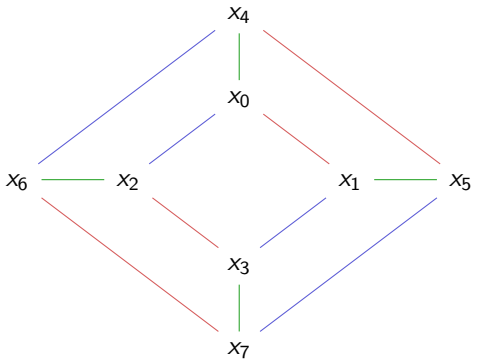


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IIIP Theorem I

IIIP has a unique solution

$\iff$ G is connected

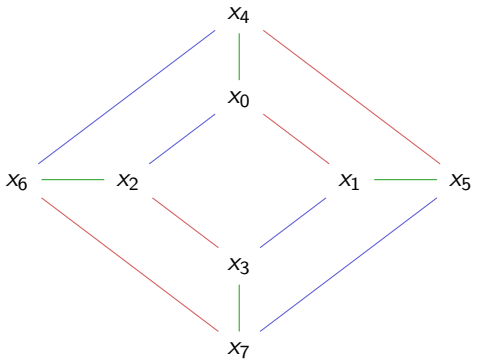
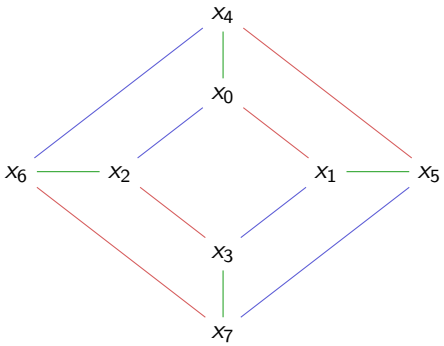


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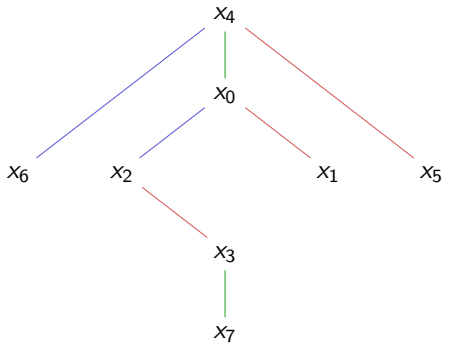
Formulae from graph theory

Let $\mathcal{T}$ be a spanning tree of G.



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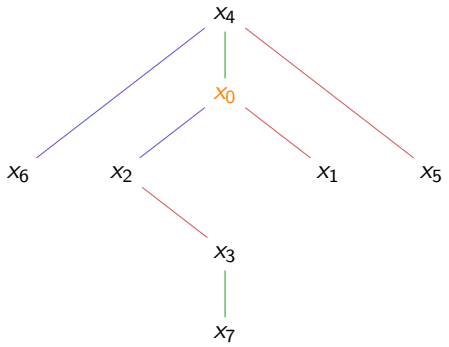
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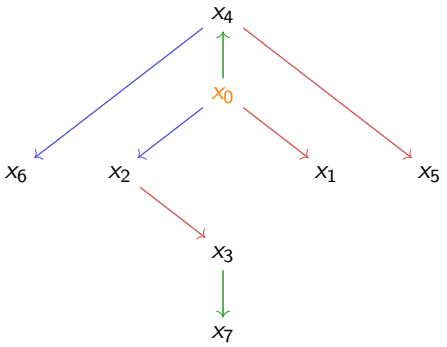
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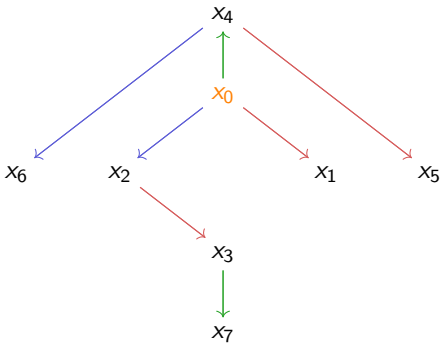


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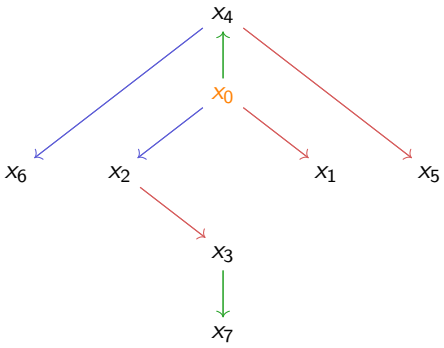
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IIIP Theorem II

This is a valid solution IIIP !



Fast formulae using Hamiltonian paths

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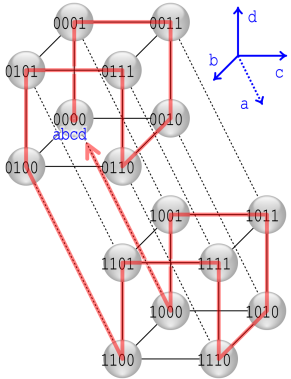


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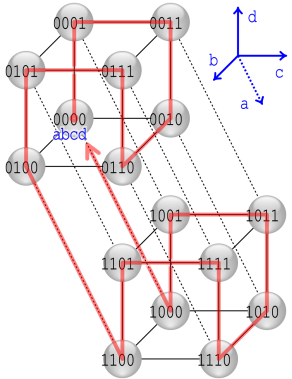


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- Total cost : $3(2^g - 2)\mathbf{M}$

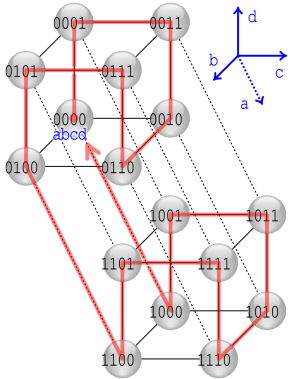


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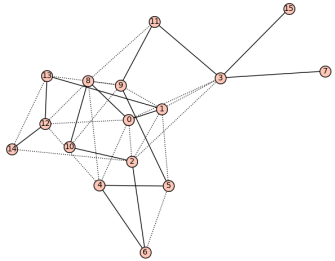


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MIKE in 1 slide

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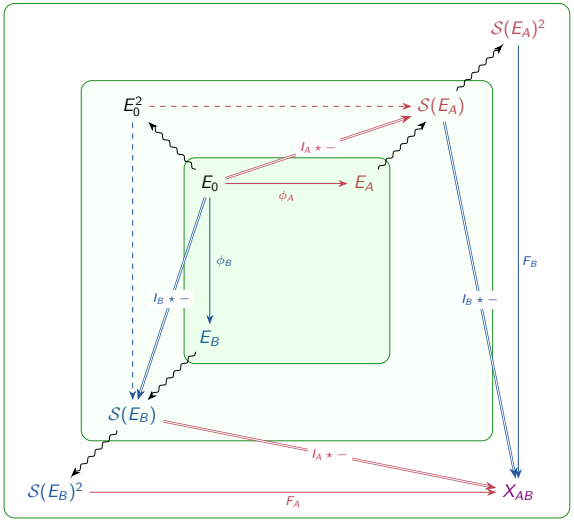
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Proof: trivial ($\simeq$ 100 pages of appendices)

MIKE (the NIKE)



MIKE

Alice		**Bob**
$s_A \in_\$ \mathbb{Z}_{2^e}$		$s_B \in_\$ \mathbb{Z}_{2^e}$
$E_A := \phi_A(E_0)$		$E_B := \phi_B(E_0)$
	$E_A \longrightarrow$	
	$\longleftarrow E_B$	
$I_A \star_R \mathcal{S}(E_B) \cong$	$(I_A \otimes I_B) \star_R E_0$	$\cong I_B \star_R \mathcal{S}(E_A)$

If you want too much details, read [BDD+26]

MIKE: 4D isogenies

- All abelian fourfold X in MIKE are of **Weil type**-$(2, 2)$.

$$\mathbb{Z}[\mathbf{i}] \hookrightarrow \mathrm{End}(X) \text{ primitively}$$

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i-compatible theta structure

For $\theta^X : A \mapsto \mathbb{P}^{15}$ induced by a **i**-compatible symplectic basis

$$\forall P \in X, \quad \theta_j^X\big([\mathbf{i}](P)\big) = \theta_{\sigma\circ j}^X(P)$$

with $\sigma = (03)(12)$.

Graph consequences

$$\theta_i^X(0_X) = \theta_{\sigma\circ i}^X(0_X)$$

IIIP over $G \iff$ IIIP over $G_{/\sim_\sigma}$

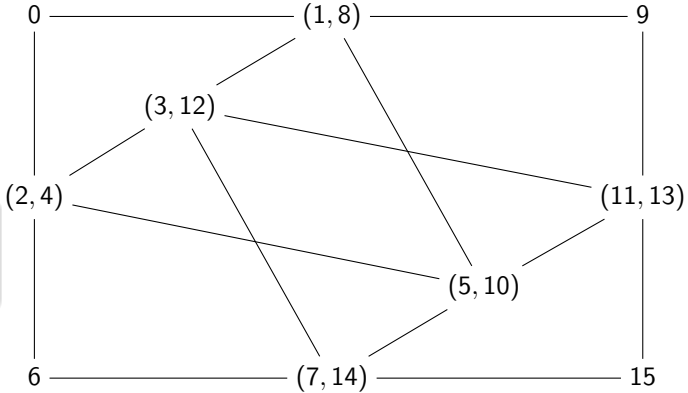
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MIKE's 4D graph

- Connected using only 2 points!



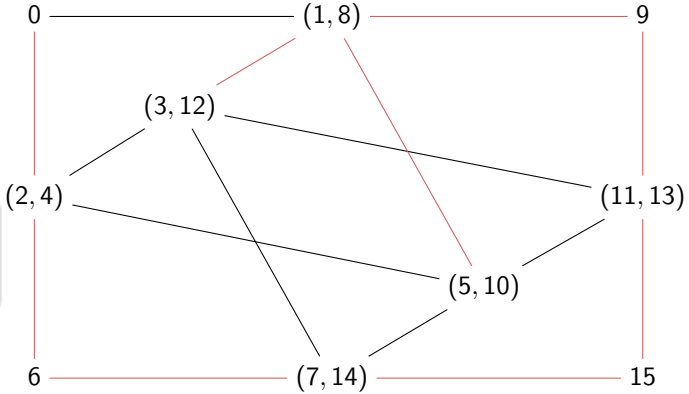
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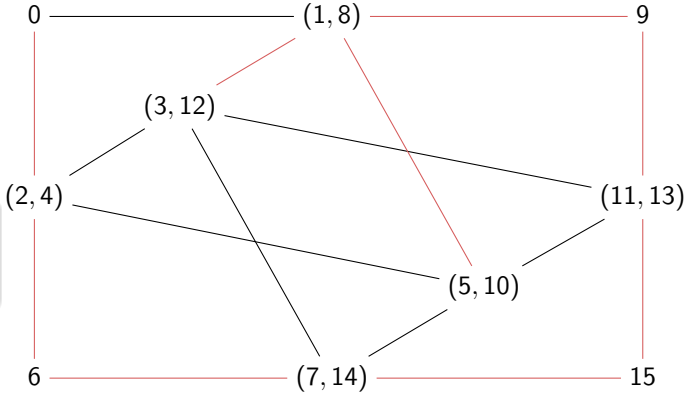
- Connected using only 2 points!
- Not Hamiltonian



Graph consequences

$$\theta_i^X(0_X) = \theta_{\sigma\circ i}^X(0_X)$$

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MIKE's 4D graph

- Connected using only 2 points!
- Not Hamiltonian

Consequence

MIKE's 4D isogeny chain is about 2x faster than generic

MIKE's timing

		Parameters / Sizes				x86 `ref`		x86 `asm`		ARM `asm`	
		p	$\log_2(p)$	e	pk B	KG MCy	SK MCy	KG MCy	SK MCy	KG MCy	SK MCy
Fast	I	$633 \cdot 2^{308} - 1$	318	228	80	3.75	27.3	2.15	16.0	1.09	8.17
	III	$593 \cdot 2^{474} - 1$	484	340	122	9.22	69.2	6.96	49.3	3.25	26.0
	V	$317 \cdot 2^{628} - 1$	637	452	160	21.9	146	13.7	94.2	6.34	50.7
Rigorous	I	$117 \cdot 2^{374} - 1$	381	372	96	8.24	61.7	4.90	34.7	2.38	17.7
	III	$77 \cdot 2^{566} - 1$	573	564	144	24.0	161	15.0	103	6.87	54.3
	V	$41 \cdot 2^{758} - 1$	764	756	192	51.8	340	36.8	230	15.5	123

Figure: Benchmarks of C implementation of MIKE [2]

[2] Timing measured on ARM M4 Pro chip, 4.5GHz.
If you wonder why 2 sets of parameters, blame Benjamin Wesolowski.

Other locus of interest: Scholten

- [Sch03] Scholten construction:

  isogenies between elliptic curve over $\mathbb{F}_{p^2}$

  $$\Downarrow$$

  isogenies between abelian surface over $\mathbb{F}_p$

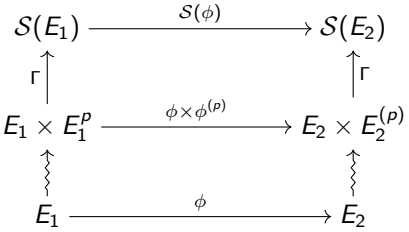
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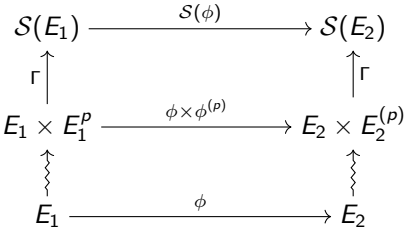
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- [Cos18, CR26]: What if the fastest way to compute a 1D isogeny was in 2D ?

Fast Scholten isogenies

The Graph PoV also applies to Weil & Scholten.

Consequences

Have fast, single point formulae for:

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	$\log_2(p)$	e	Time
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Table: Speed comparison on Scholten computing

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Table: Speed comparison on Scholten computing

- Costello dream is within reach.
- But still have open questions to solve:
  - = gluing/spitting problems.

Conclusion

Seeing isogenies as graph is a powerful tool.

1. The right way to see the problem!

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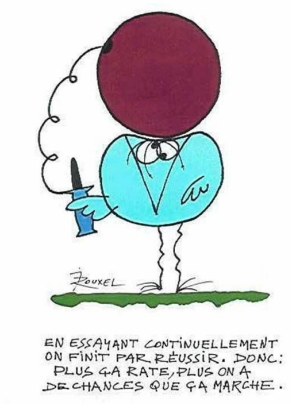
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Thanks for listening !

Do you have questions ?

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